15 · Solution Step 5: Three Losses, Three Kinds of Mistake

1. Goal of this step

Define "the student is doing well" precisely enough to train on. One number, the **loss**, must go down when the student gets better.

2. What the paper does

It combines three losses, each measuring a different mistake:

Loss	Question it asks	Plain-words version	Math file
Alignment $\mathcal{L}_{\text{align}}$	Does the student point where its teacher points?	"Copy your teacher."	`math/08`
Contrastive $\mathcal{L}_{\text{contra}}$ (InfoNCE)	Is the student closer to *its own* teacher than to the other teachers in the batch?	"Don't be confusable with other queries."	`math/09`
Rank distillation $\mathcal{L}_{\text{rank}}$	Does the student put the teacher's top documents in the teacher's order?	"Rank documents like your teacher."	`math/10`

$$\mathcal{L} = \lambda_a \mathcal{L}_{\text{align}} + \lambda_c \mathcal{L}_{\text{contra}} + \lambda_r \mathcal{L}_{\text{rank}}$$

The best setting uses $\lambda_a = \lambda_c = \lambda_r = 0.5$, temperatures $\tau = \tau_r = 0.05$, and the teacher's top $K = 128$ documents for ranking. (`math/11` combines them.)

3. Why three and not one

The worked example shows the reason perfectly. For the query "cat food":

- **Alignment says 0.04**, nearly perfect. The student's cosine with its teacher is 0.96.
- **Contrastive says** the "cat toys" student is almost as close to the "cat food" teacher (0.936) as to its own (0.96). The two queries are nearly confusable.
- **Rank says** the student ranks *Pizza recipes* above *Cat behavior guide*, the teacher's second choice. A cosine of 0.96 is not enough to get the order right.

A vector can be *close* to the teacher and still be *wrong in the ways that matter for search*. Each loss catches a different way of being wrong.

Loss	Kind of signal	Strength	Weakness
Alignment	Absolute: "go to this point"	Strong, stable, always has a clear target	Doesn't care about neighbors or order
Contrastive	Relative to other *queries*	Keeps queries separable	Depends on what else is in the batch (only 16 queries)
Rank	Relative among *documents*	Directly targets search order	Has no anchor: collapses alone (see below)

4. What the experiments say about them

From the ablation (all at 30 epochs, lr 3×10^{-4}), Recall@10:

Setting	R@10
Align only	0.567
Contrastive only	0.498
Rank only	**0.001** (collapsed)
Align + Contrastive	0.582
Align + Rank	0.579
All three	0.595

- Alignment is the backbone.
- Each extra loss helps a bit on top of it; all three together are best.
- Rank distillation alone fails completely. It only compares scores *among* a few documents, so it never tells the vector *where in the space* to be. It can match the teacher's order among those 128 documents while pointing somewhere that retrieves junk from the rest of the index. `math/10` shows this happening with numbers.
- Too much rank weight also hurts ($\lambda_r = 1.0$: 0.516).

5. What could go wrong

- **Weights are hand-tuned.** The best λs were found by trial on this dataset.
- **Temperature matters.** Too low and training is harsh and unstable; too high and near-ties are ignored (`math/09`).
- **Batch size limits contrastive learning.** With 16 queries per batch, each student only sees 15 "wrong" teachers.